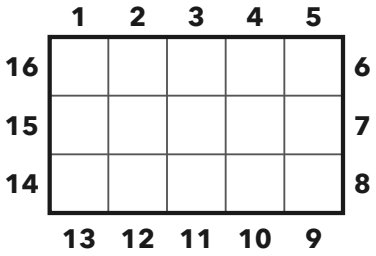
Perimeter

Perimeter is the distance all the way around.

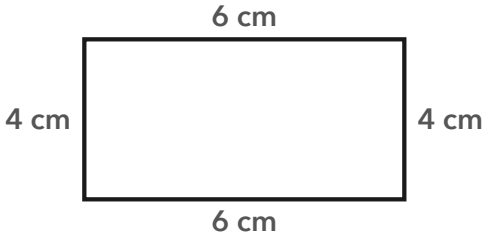
Walk around the edge and count



Perimeter = 16 units

Count each edge of a square once.

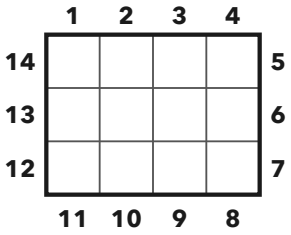
Add the sides



6 + 4 + 6 + 4 = 20 cm

Opposite sides are the same length.

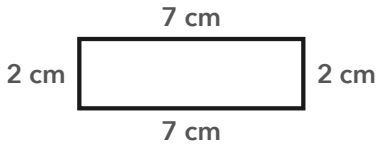
Area is inside. Perimeter is around.



Area = 12 square units

Perimeter = 14 units

Your turn!



7 + 2 + 7 + 2 = 18 cm

For teachers

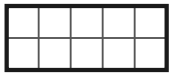
Goal: perimeter is the length around a shape (3.MD.8). Trace the edge with a finger while counting. Then add the side lengths. Keep asking "inside or around?" so area and perimeter don't get mixed up.

Perimeter: Practice

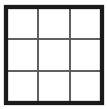
Go all the way around. Add every side.

Part 1 Count around Count every edge on the outside.

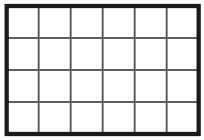
1) **Perimeter = 14 units**



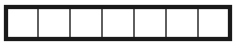
2) **Perimeter = 12 units**



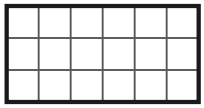
3) **Perimeter = 20 units**



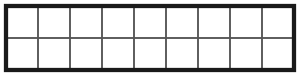
4) **Perimeter = 16 units**



5) **Perimeter = 18 units**

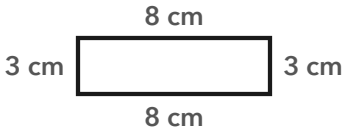


6) **Perimeter = 22 units**

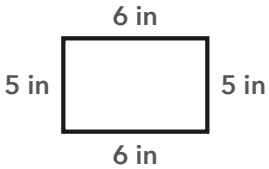


Part 2 Add the sides Write each side, then add.

7) 8 + 3 + 8 + 3 = 22 cm



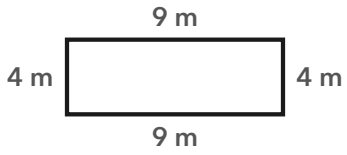
8) 6 + 5 + 6 + 5 = 22 in



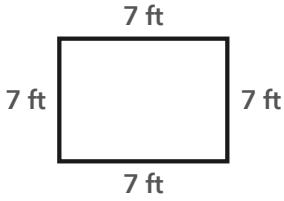
Perimeter: Practice (continued)

Part 2 Add the sides (continued)

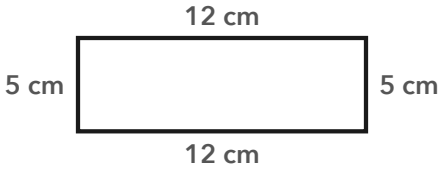
9) 9 + 4 + 9 + 4 = 26 m



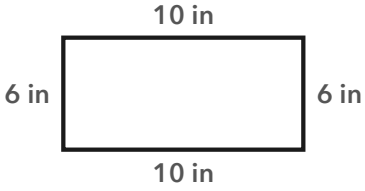
10) 7 + 7 + 7 + 7 = 28 ft



11) 12 + 5 + 12 + 5 = 34 cm

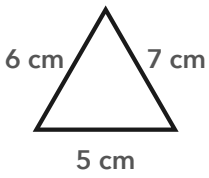


12) 10 + 6 + 10 + 6 = 32 in

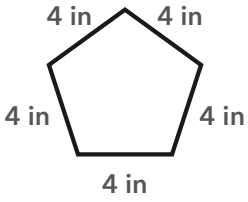


Part 3 More sides Add all the sides.

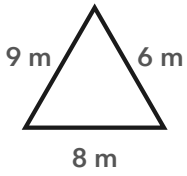
13) 5 + 7 + 6 = 18 cm



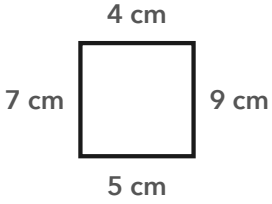
14) 4 + 4 + 4 + 4 + 4 = 20 in



15) 8 + 6 + 9 = 23 m



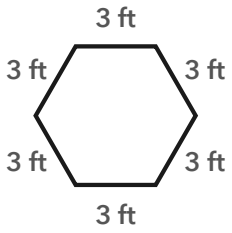
16) 5 + 9 + 4 + 7 = 25 cm



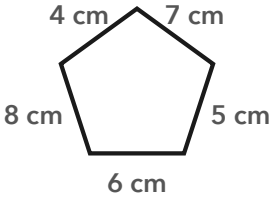
Perimeter: Practice (continued)

Part 3 More sides (continued)

17) $3 + 3 + 3 + 3 + 3 + 3 =$ 18 ft

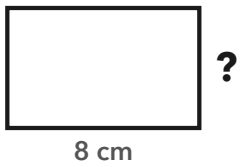


18) $6 + 5 + 7 + 4 + 8 =$ 30 cm



Part 4 Find the missing side The perimeter is given. Take away the sides you know.

19) **Perimeter = 26 cm. Find the missing side.**



$8 + 8 =$ 16

$26 -$ 16 $=$ 10

half of 10 $=$ 5 cm

20) **Perimeter = 18 in. Find the missing side.**

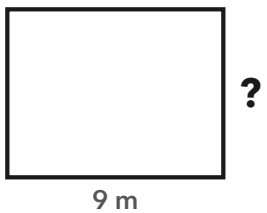


$6 + 6 =$ 12

$18 -$ 12 $=$ 6

half of 6 $=$ 3 in

21) **Perimeter = 32 m. Find the missing side.**



$9 + 9 =$ 18

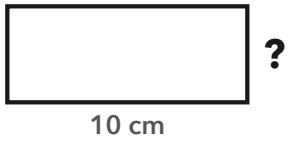
$32 -$ 18 $=$ 14

half of 14 $=$ 7 m

Perimeter: Practice (continued)

Part 4 Find the missing side (continued)

22) **Perimeter = 28 cm. Find the missing side.**



10 + 10 = 20

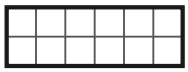
28 − 20 = 8

half of 8 = 4 cm

Part 5 Area and perimeter Inside, then around.

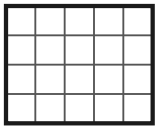
23) Area = 12 square units

Perimeter = 16 units



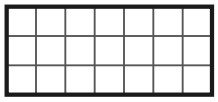
24) Area = 20 square units

Perimeter = 18 units



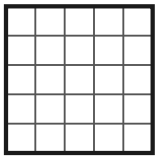
25) Area = 21 square units

Perimeter = 20 units



26) Area = 25 square units

Perimeter = 20 units



Part 6 Rectangles No help this time. Add all four sides.

27) 5 cm by 3 cm: perimeter = 16 cm

28) 9 cm by 2 cm: perimeter = 22 cm

29) 7 cm by 4 cm: perimeter = 22 cm

30) 12 cm by 6 cm: perimeter = 36 cm

Perimeter: Practice (continued)

Part 6 Rectangles (continued)

31) **11 cm by 3 cm: perimeter =** 28 **cm**

32) **6 cm by 4 cm: perimeter =** 20 **cm**

33) **15 cm by 5 cm: perimeter =** 40 **cm**

34) **10 cm by 7 cm: perimeter =** 34 **cm**

35) **9 cm by 6 cm: perimeter =** 30 **cm**

36) **14 cm by 2 cm: perimeter =** 32 **cm**

Part 7 Squares No help this time. All four sides are the same.

37) **5 m by 5 m square: perimeter =** 20 **m**

38) **8 in by 8 in square: perimeter =** 32 **in**

39) **3 cm by 3 cm square: perimeter =** 12 **cm**

40) **10 ft by 10 ft square: perimeter =** 40 **ft**

41) **6 m by 6 m square: perimeter =** 24 **m**

42) **9 cm by 9 cm square: perimeter =** 36 **cm**

Perimeter: Practice (continued)

Part 8 Add the sides No help this time.

43) $15 + 8 + 15 + 8 =$ **46**

44) $7 + 9 + 11 =$ **27**

45) $13 + 6 + 13 + 6 =$ **38**

46) $12 + 12 + 12 =$ **36**

47) $9 + 14 + 9 + 14 =$ **46**

48) $6 + 8 + 5 + 9 =$ **28**

Part 9 Find the missing side No help this time. Rectangles only.

49) 7 cm by **3** cm: perimeter = 20 cm

50) 5 in by **4** in: perimeter = 18 in

51) 9 m by **2** m: perimeter = 22 m

52) 8 ft by **6** ft: perimeter = 28 ft

53) 11 cm by **4** cm: perimeter = 30 cm

54) 6 in by **5** in: perimeter = 22 in

Perimeter: Practice (continued)

Part 10 Find the perimeter

55) 3 cm by 6 cm: perimeter = 18 cm

56) 10 ft by 3 ft: perimeter = 26 ft

57) 3 in by 11 in: perimeter = 28 in

58) 11 m by 9 m: perimeter = 40 m

59) 3 in by 3 in: perimeter = 12 in

60) 6 m by 11 m: perimeter = 34 m

61) 3 m by 3 m: perimeter = 12 m

62) 4 in by 8 in: perimeter = 24 in

63) 2 cm by 2 cm: perimeter = 8 cm

64) 8 cm by 3 cm: perimeter = 22 cm

65) 15 m by 2 m: perimeter = 34 m

66) 2 ft by 9 ft: perimeter = 22 ft

Perimeter: Practice (continued)

Part 10 Find the perimeter (continued)

67) 11 ft by 11 ft: perimeter = 44 ft

68) 4 m by 6 m: perimeter = 20 m

69) 11 ft by 7 ft: perimeter = 36 ft

70) 2 ft by 8 ft: perimeter = 20 ft

71) 5 m by 10 m: perimeter = 30 m

72) 11 cm by 2 cm: perimeter = 26 cm

73) 5 ft by 5 ft: perimeter = 20 ft

74) 10 in by 9 in: perimeter = 38 in

75) 12 m by 7 m: perimeter = 38 m

76) 7 in by 2 in: perimeter = 18 in

77) 3 ft by 6 ft: perimeter = 18 ft

78) 7 ft by 4 ft: perimeter = 22 ft

Perimeter: Practice (continued)

Part 11 Find the missing side

79) 11 cm by [2] cm: perimeter = 26 cm

80) 14 m by [9] m: perimeter = 46 m

81) 15 m by [2] m: perimeter = 34 m

82) 12 m by [6] m: perimeter = 36 m

83) 15 m by [4] m: perimeter = 38 m

84) 7 m by [3] m: perimeter = 20 m

85) 6 cm by [9] cm: perimeter = 30 cm

86) 5 m by [5] m: perimeter = 20 m

87) 12 m by [7] m: perimeter = 38 m

88) 14 m by [4] m: perimeter = 36 m

89) 9 m by [7] m: perimeter = 32 m

90) 13 cm by [4] cm: perimeter = 34 cm

Perimeter: Practice (continued)

Part 11 Find the missing side (continued)

91) 9 m by [10] m: perimeter = 38 m

92) 12 cm by [4] cm: perimeter = 32 cm

93) 13 cm by [12] cm: perimeter = 50 cm

94) 12 m by [4] m: perimeter = 32 m

95) 14 cm by [12] cm: perimeter = 52 cm

96) 6 cm by [6] cm: perimeter = 24 cm

Part 12 Remember? Problems from earlier packets.

97) $9 \times 30 =$ [270]

98) Round 452 to the nearest ten: [450]

99) 5 cm by 7 cm: area = [35] sq cm

100) 8 eighths = [1] whole

101) $500 - 238 =$ [262]

102) 8 groups of 7 = [56]

Perimeter: Practice (continued)

Part 13 Word problems Draw the shape and label the sides.

103) A garden is 9 feet long and 5 feet wide. How much fence goes all the way around it?

Number sentence **9 + 5 + 9 + 5 = 28**

Answer **28** feet

Draw or show your work

104) A square picture frame has sides of 9 inches. What is its perimeter?

Number sentence **9 + 9 + 9 + 9 = 36**

Answer **36** inches

Draw or show your work

105) A triangle has sides of 6 cm, 8 cm and 10 cm. What is its perimeter?

Number sentence **6 + 8 + 10 = 24**

Answer **24** cm

Draw or show your work

106) A rug is 6 feet long. Its perimeter is 20 feet. How wide is it?

Number sentence **20 − 12 = 8, half of 8 = 4**

Answer **4** feet

Draw or show your work

107) Challenge: A playground is 10 m long and 8 m wide. Kim walks around it 2 times. How far does Kim walk?

Number sentence **10 + 8 + 10 + 8 = 36, 36 + 36 = 72**

Answer **72** m

Draw or show your work

Check-In: Perimeter

Show what you know. No help this time.

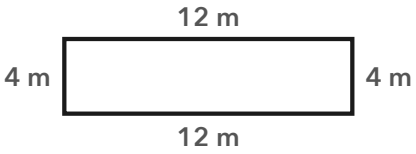
Mastery: 8 of 10 (80%)

Part 1 Add the sides

1) 12 + 3 + 12 + 3 = 30 m



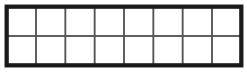
2) 12 + 4 + 12 + 4 = 32 m



Part 2 Area and perimeter

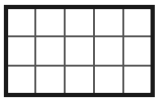
3) Area = 16 square units

Perimeter = 20 units



4) Area = 15 square units

Perimeter = 16 units



Part 3 Perimeter

5) 8 m by 5 m: perimeter = 26 m

6) 11 in by 5 in: perimeter = 32 in

7) 7 cm by 6 cm: perimeter = 26 cm

8) 11 cm by 5 cm: perimeter = 32 cm

Part 4 Find the missing side

9) 5 cm by 4 cm: perimeter = 18 cm

10) 10 cm by 5 cm: perimeter = 30 cm